Distribution of Effective Category Counts

All decoder MLP neurons; 1,000 samples per run; $K_{eff} = \exp(S) = 4^{H_{norm}}$

Smoothed density Histogram (bin = 0.05)

	MME	MMBench	TextVQA	ChartQA
LLaVA-1.5-7B	Median 2.21 IQR 1.74–2.85	Median 2.30 IQR 1.81–2.94	Median 2.38 IQR 1.91–3.04	Median 2.00 IQR 1.69–2.65
Qwen3-VL-2B	Median 1.94 IQR 1.42–2.50	Median 2.16 IQR 1.60–2.81	Median 1.94 IQR 1.43–2.53	Median 1.99 IQR 1.43–2.66
Qwen3.5-2B	Median 2.08 IQR 1.55–2.72	Median 2.43 IQR 1.83–3.07	Median 2.15 IQR 1.66–2.85	Median 2.17 IQR 1.61–2.93
Gemma-4-E2B-it	Median 1.89 IQR 1.37–2.45	Median 1.99 IQR 1.49–2.60	Median 2.01 IQR 1.49–2.63	Median 1.98 IQR 1.46–2.61

Neurons per bin (%)

Effective category count K_{eff}